

— IQOO HACKATHON 2026 · PUNE CITY BATTLE · WORKING PROFESSIONALS

CLASHFIT

Your body is the controller.
Your camera is the referee.

Camera-verified effort becomes damage against a boss. Pose estimation, form scoring, a fatigue model and a language model all run on the phone — and nothing ever leaves it.

TEAM DA GOATS

OMKAR KADAM

UJJWAL PARDESHI

WORKING PROTOTYPE RUNNING

CLASH-FIT.VERCEL.APP

— THE PROBLEM

NOTHING ON YOUR PHONE CAN SEE THE REP.

Every fitness app counts what you **tell** it. Tap ten, it logs ten. There is no sensor in that loop — only a text field and your honesty at the exact moment you are least able to judge yourself.

WHAT THE APP KNOWS

"10 squats"

A number you typed. No depth, no range, no tempo, no alignment, no idea whether rep 9 looked anything like rep 1.

WHAT ACTUALLY HAPPENED

10 reps. 4 of them shallow.

Depth fell 4 cm across the set. Descent got faster as control went. The knee drifted inside the toe on the last three.

The gap between those two boxes is the entire product.

WHY IT MATTERS

A LOG THAT CANNOT BE WRONG IS A LOG THAT CANNOT HELP.

01

The number goes up while the movement falls apart

Self-reported counts drift toward whatever keeps the streak alive. The log rewards the tap, not the rep, so the two quietly stop describing the same thing.

02

Nobody tells you when to stop

Training effect and injury risk both live in the last few reps of a set. Without a measure of fatigue there is no way to know which of those two you are buying.

03

Form coaching needs eyes, and eyes cost money

The feedback that fixes a movement is the one thing a phone app has never been able to give — so it is priced at whatever an hour of a coach's time costs.

**“The problem was never motivation.
It is that nobody is watching.”**

Which is a sensing problem, not a behaviour-change problem — and sensing problems are the kind you can actually solve with a camera and a model that fits on a phone.

— THE INSIGHT

A CAMERA THAT CAN SEE A BODY IS A REFEREE.

Once every rep is graded before it counts, effort stops being something you claim and becomes something you spend. That single change is what turns training into a game that cannot be cheated.

— HOW IT WORKS

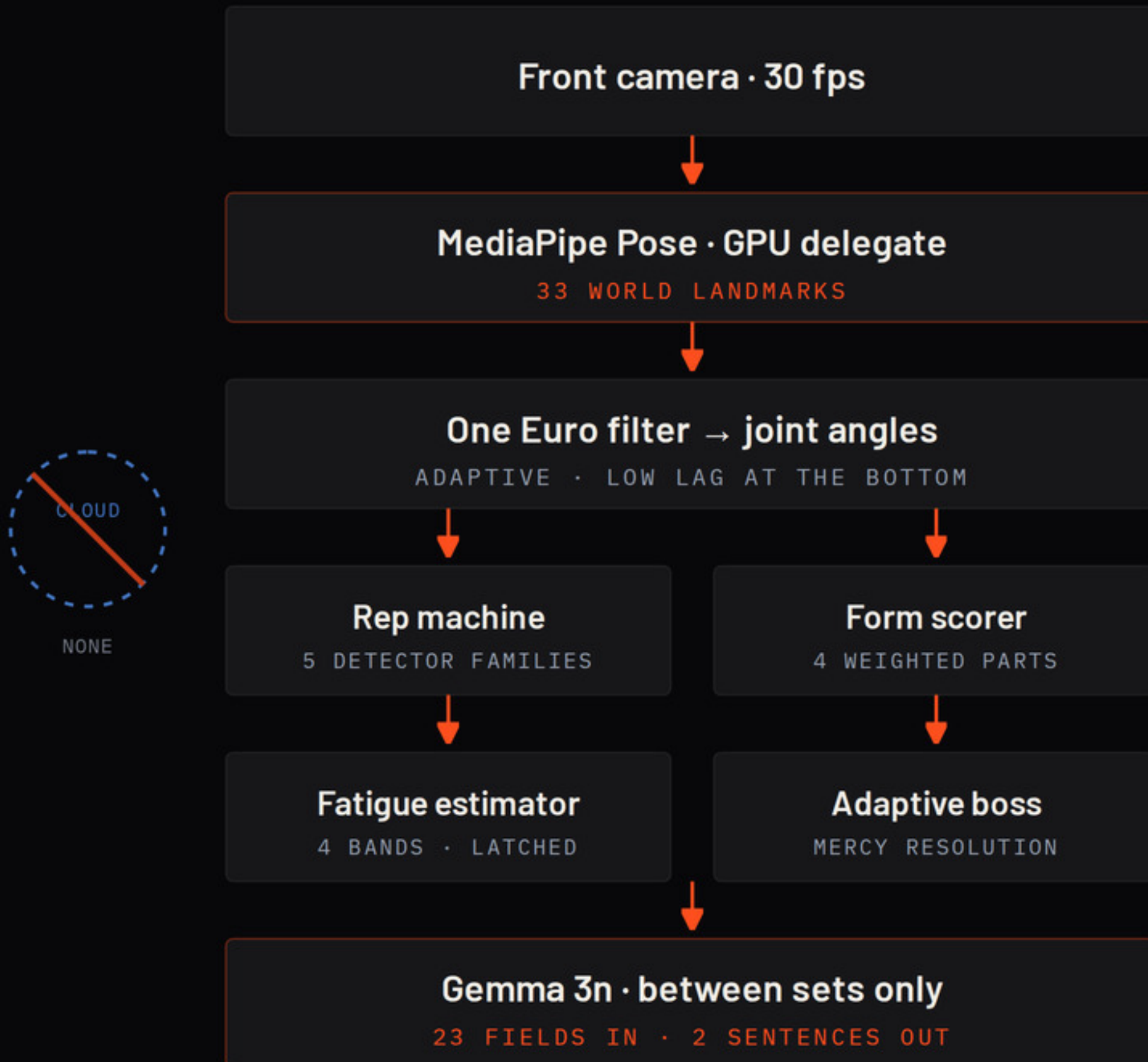
REPS IN. DAMAGE OUT.

EVERY BOX BELOW RUNS ON THE PHONE. THERE IS NO SERVER IN THIS DIAGRAM.



A half-rep buys nothing. That is the whole design: the only way to do more damage is to move better, and the phone is the thing deciding whether you did.

NO CLOUD BOX ON THIS DIAGRAM.



GPU

Pose runs on the GPU delegate

Which leaves the CPU free for the rep machine, the fatigue model and the fight, in the same frame budget.

CPU

Everything between is arithmetic

Filters, angles, a state machine and four weighted scores. Deterministic, testable, and fast enough that it never becomes the bottleneck.

NPU

The language model runs between sets

Never while the camera does. The phone does one hard thing at a time, on purpose.

NET

No internet permission at all

Not a policy, a build property. There is nothing in the app that could upload a frame even if it wanted to.

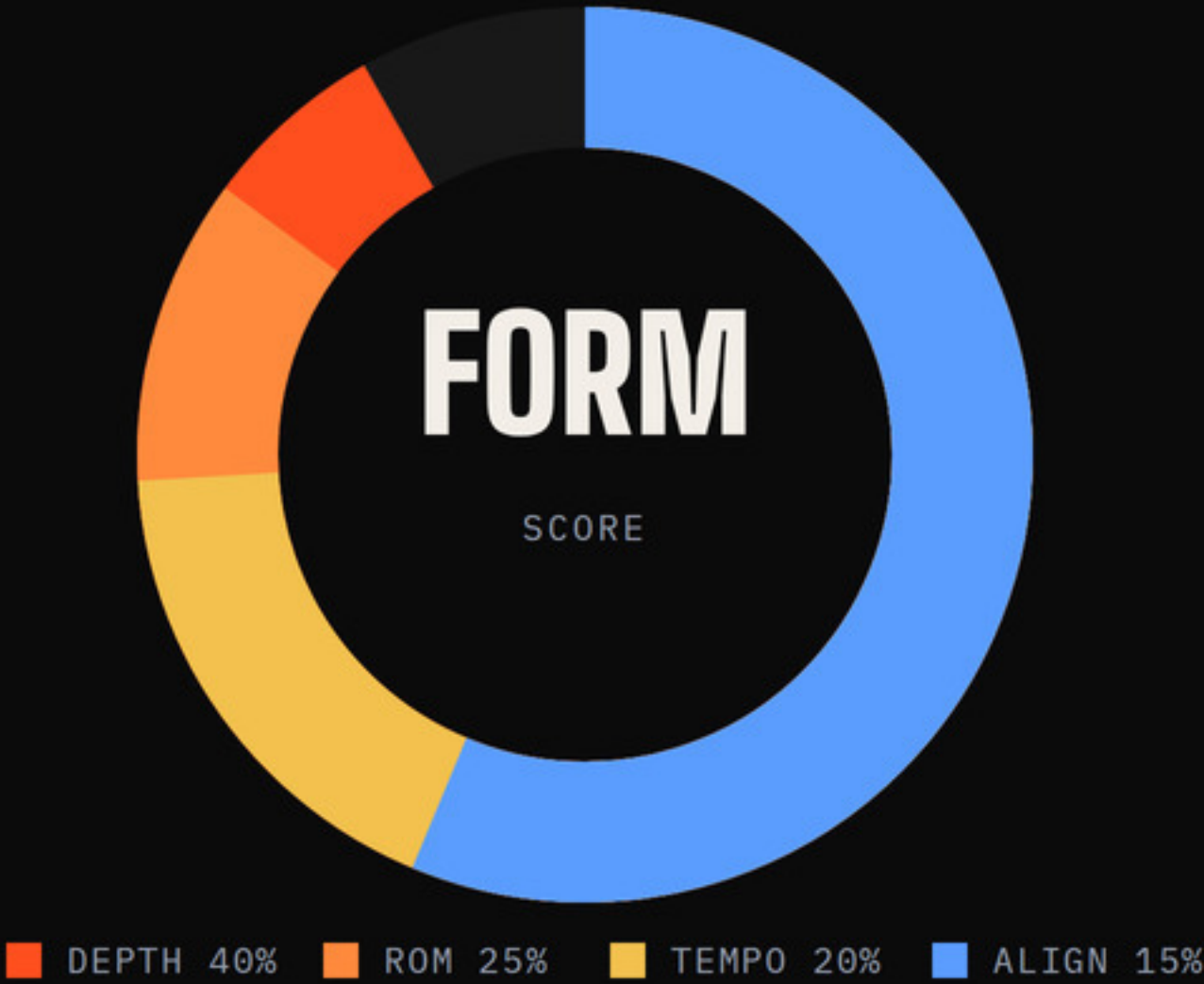
— WHAT “GRADED” MEANS

ONE REP, FOUR MEASUREMENTS.

A rep is not a count. It is a weighted score over four quantities the camera can actually measure — and the weights differ per exercise, because depth matters more in a squat than tempo does.

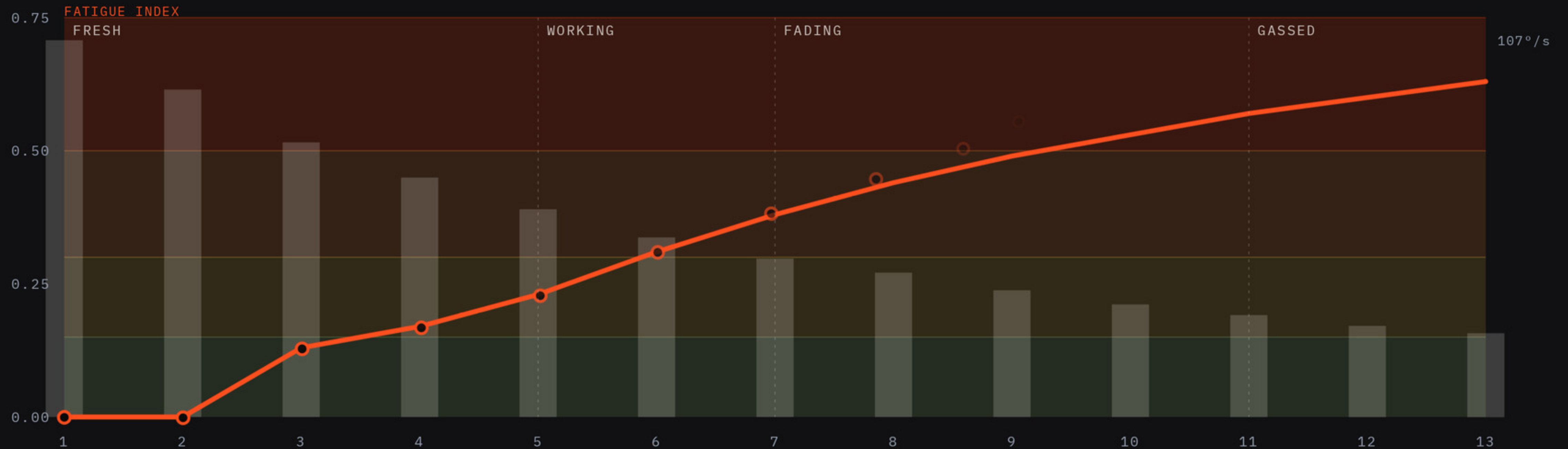
WEIGHT	WHAT IT MEASURES
0.40	Depth — how far the joint actually travelled, in real centimetres, scored superlinearly so a shallow rep loses more than it looks like it should
0.25	Range — the arc of this rep against the best arc you produced in your own first three
0.20	Tempo — the eccentric, because dropping into the bottom is how a rep gets cheap and how people get hurt
0.15	Alignment — the joint that betrays the movement: knee over toe, hip above shoulder, elbow flare

SQUAT WEIGHTS, FROM CONFIG/EXERCISES/SQUAT.JSON



IT KNOWS WHEN YOU'RE GASSED.

Velocity, range and the pauses between reps, measured every rep against **your own first three**. Not a difficulty slider. Not a self-reported effort rating. The point at which the movement actually changed.



— ONE MODEL, FIVE SHAPES OF MOVEMENT

A PLANK HAS NO VELOCITY. SO WE MEASURE THE SHAKE.

Fatigue is not one signal. Each family of movement degrades in its own way, so each has its own weighted signals feeding the same estimator and the same four bands. That is what lets one engine cover yoga, cardio, holds, jumps and reps.

FAMILY	EXERCISES	GAME	HOW FATIGUE SHOWS UP
REP_CYCLE	10	Boss Fight · strength reps	Velocity decay 0.45 · range decay 0.35 · growing pauses 0.20
ISOMETRIC_HOLD	9	Siege · your plank is the shield	Tremor growth 0.60 · quality decay 0.40
POSE_MATCH	14	Sigil · each asana a constellation	Accuracy drift 1.00
CADENCE	11	Pursuit · distance is cadence	Cadence decay 0.55 · amplitude decay 0.45
BALLISTIC	7	Breaker · jump height in centimetres	Height decay 0.60 · landing softness 0.40

51

EXERCISES

5

DETECTOR FAMILIES

1

FATIGUE ESTIMATOR

4

BANDS · 0.15 / 0.30 / 0.50

A BAD REP VISIBLY DOES LESS.

Damage is the form score raised to an exponent, multiplied by a combo that only clean reps extend. You can feel the difference between a good rep and a lazy one in the number that comes off the boss — which is the entire coaching mechanism, delivered without a word.



The boss reads your fatigue band

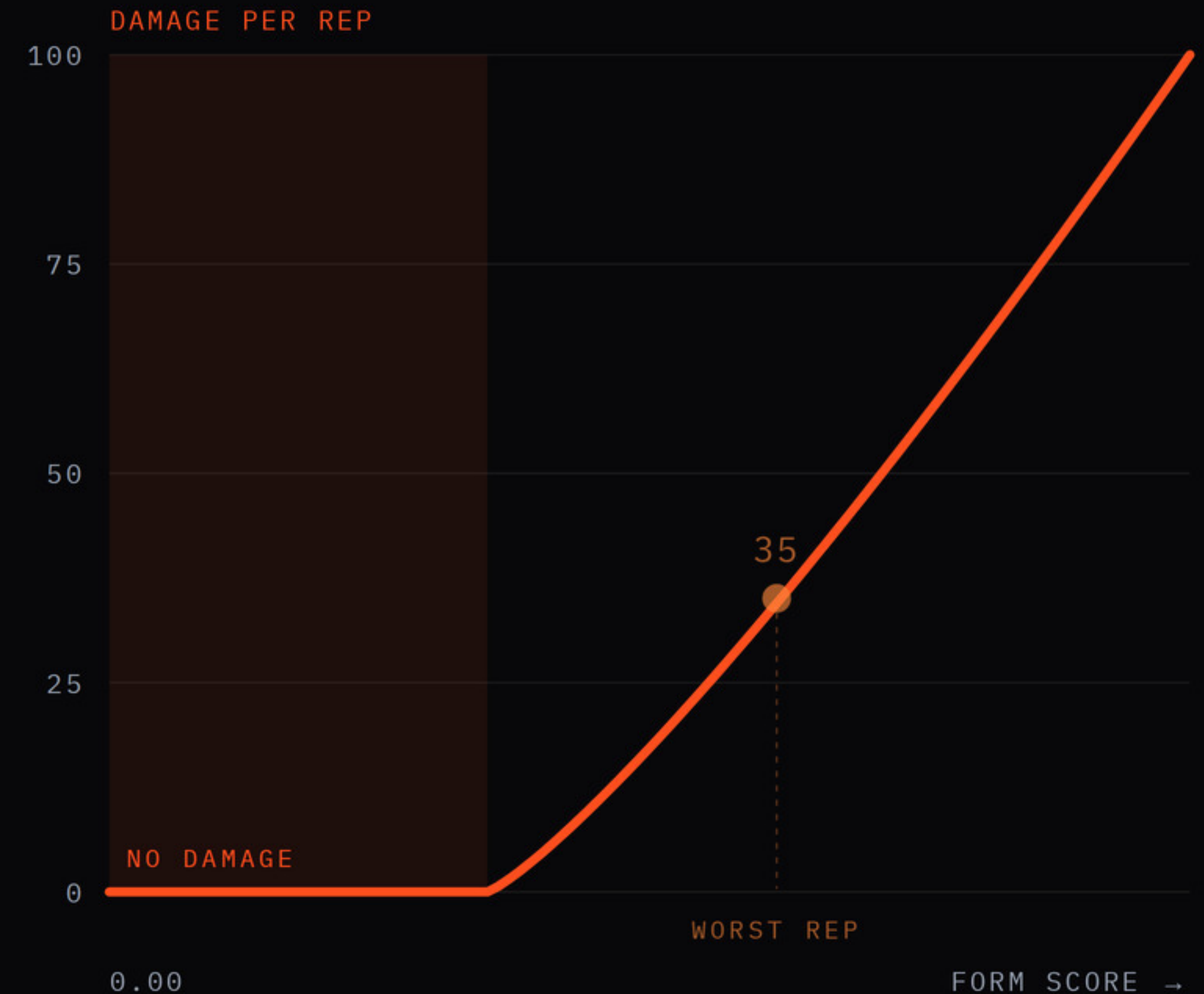
It staggers when you fade rather than scaling difficulty behind your back.



Mercy resolution

When you are genuinely done, the fight ends. A game that punishes an honest athlete for reaching failure is a game nobody opens twice.

“The boss can't outlast you —
but it makes you earn the ending.”



THE COACH AND THE VILLAIN ARE THE SAME MODEL.

Between sets — never during one — a twenty-three-field summary of what your body just did goes to **Gemma 3n**, running on the phone. It writes the coaching and it writes the taunt. It never sees a frame of video, because we never send it one.

GOES TO THE MODEL	NEVER GOES
Rep count, form means, first-three vs last-three	Video
Depth in centimetres, and how much of it you lost	Images
Velocity loss, range loss, fatigue band	Landmarks
Best and worst rep, combo, boss HP, rest seconds	Audio
Bilateral deficit, and which side is weaker	Identity

EVERY FIELD ENUMERATED IN SRC/COACH.JS · SUMMARISE()

“Your last three reps lost four centimetres of depth. Take forty seconds, then finish it.”

COACH

“Your knees are negotiating. I don't negotiate.”

THE PACEMAKER

And when the model is wrong, it does not speak. A validator checks every sentence against the telemetry it was given. A line that cites a number we did not measure is dropped, and a template bank keyed on the same numbers speaks instead. The athlete never hears a hallucination.

— PRIVACY

A CAMERA IN YOUR BEDROOM THAT CANNOT PHONE HOME.

Pose, form scoring, fatigue and the language model all run on the phone. The shipping app requests no internet permission at all — so the promise is not a policy you have to believe, it is a property you can check.

0

BYTES UPLOADED

0

ACCOUNTS

0

FRAMES STORED

0

RUNTIME DEPENDENCIES

— WHY THIS PHONE

TWO HARD JOBS, ONE BATTERY.

90°

The spec that actually decides it

The 32 MP front camera's ninety-degree field of view holds a whole standing body from about two metres — the distance a bedroom gives you. A narrower selfie lens crops your feet, and a rep the camera cannot see is never counted.

3 nm

Snapdragon 8 Elite Gen 5

Pose on the GPU delegate leaves the CPU free for the rep machine, the fatigue model and the fight inside one frame's budget — then the same silicon runs a language model between sets with the radio off.

7000

A set is not a benchmark run

Twelve minutes of continuous camera, GPU and screen is a thermal problem, not a peak one. A 7000 mAh cell and iQOO's 8K vapour-chamber cooling are what keep the frame rate from sagging halfway through and taking the fatigue baseline with it.

× 2

Two of them make a duel

The multiplayer story needs a second phone and nothing else — no server, no router, no account.



SPECIFICATIONS CROSS-VERIFIED AGAINST IQOO'S OWN PRODUCT PAGES, GSMARENA AND THE CHIP VENDOR. NOTHING ON THIS SLIDE IS QUOTED FROM A SPEC WE COULD NOT SOURCE TWICE.

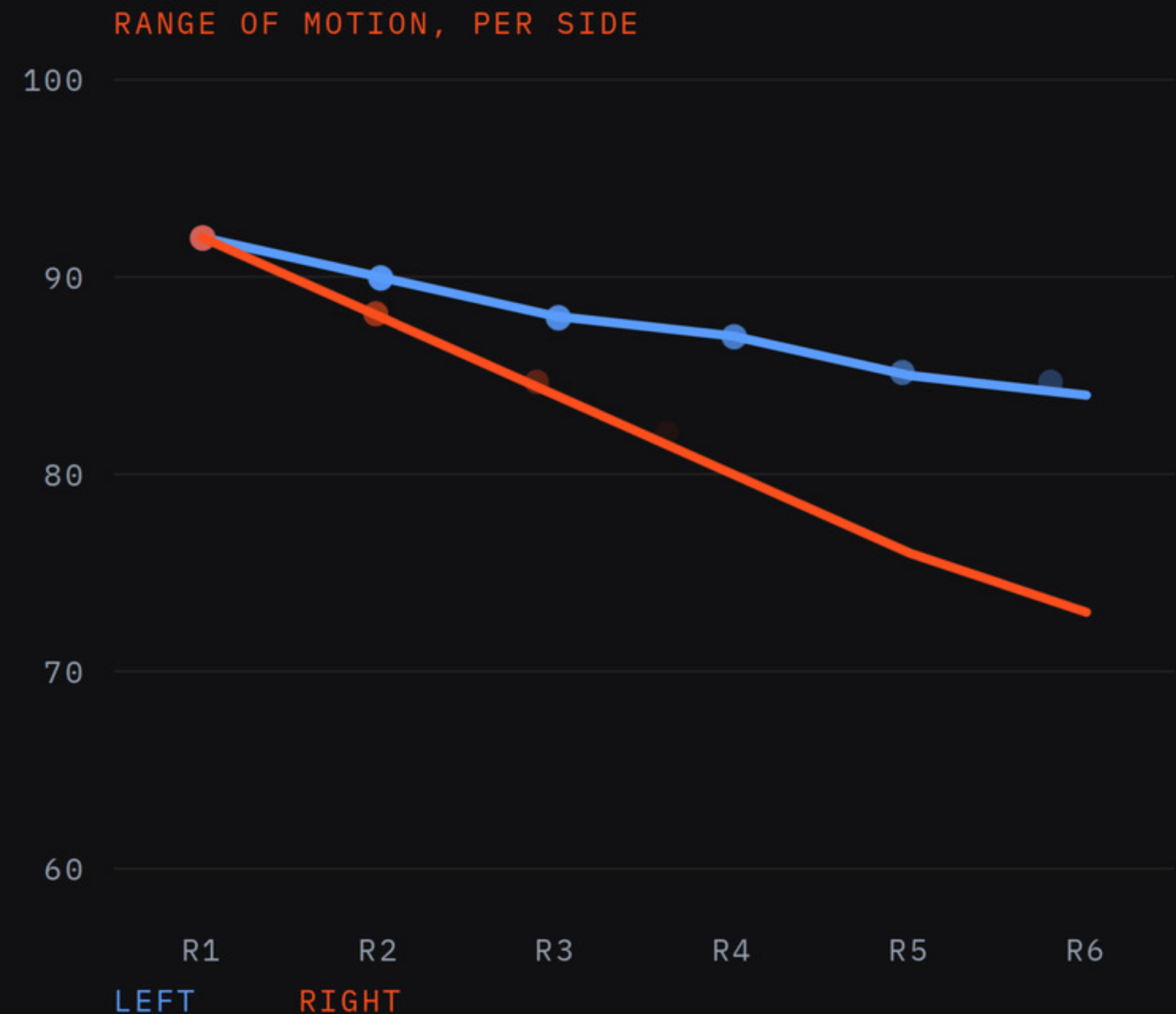
BOTH SIDES WERE ALWAYS THERE. WE STOPPED AVERAGING THEM AWAY.

Pose gives you a left knee and a right knee. Every app averages them into one number and throws the difference away. That difference — the Limb Symmetry Index — is a measure clinicians actually use, and no consumer fitness app reports it, because none of them can see both sides.

$$\text{LSI} = \text{weaker} / \text{stronger} \times 100$$

Computed per rep across a set, then graded for confidence — because a side-on camera foreshortens the far limb and will manufacture an asymmetry that is not there. When the framing cannot support the number, we say so instead of reporting it.

The output is a measurement and a phrase. Never a diagnosis: the language validator will not let the words **injury**, **risk** or **abnormal** reach the athlete.



THE TESTS A PHYSIO USES, ON A PHONE YOU ALREADY OWN.

The 30-second sit-to-stand is a published, validated test of lower-limb strength. It is counted by a human with a stopwatch, in a clinic, once or twice a year. The camera can count it correctly, at home, as often as you like — and plot the trend that the twice-a-year version can never show.

01 Counted the way the protocol defines it

Full stand, full sit, thirty seconds, no arms. The same rules the paper specifies.

02 A five-month trend, not a single number

The value of a home test is the slope, which is exactly the thing an annual assessment cannot give you.

03 It is a measurement, not a diagnosis

Stated on the screen, in the copy, and enforced by the same validator that governs the coach's language.

14

STANDS IN 30 SECONDS

Ageing populations, post-operative rehab, and anyone whose physiotherapist wants adherence data between appointments. The same engine that runs a boss fight runs this — which is the argument for building the game first and the clinic second, rather than the other way round.

NOT A MEDICAL DEVICE

ON-DEVICE

NO ACCOUNT

TWO PHONES, ONE BOSS, NO SERVER.

Phones exchange **events**, not state. Boss HP is a pure function of the deduplicated union of every damage event either phone has seen, so the two screens converge without any authority deciding who is right.

$$\text{bossHp} = \text{maxHp} - \sum \text{damage over dedupe}(\text{union})$$

Every message carries a self-healing tail of the last eight events, so a dropped packet repairs itself on the next one. No acknowledgements, no retransmit logic, no clock sync — and it works in airplane mode.

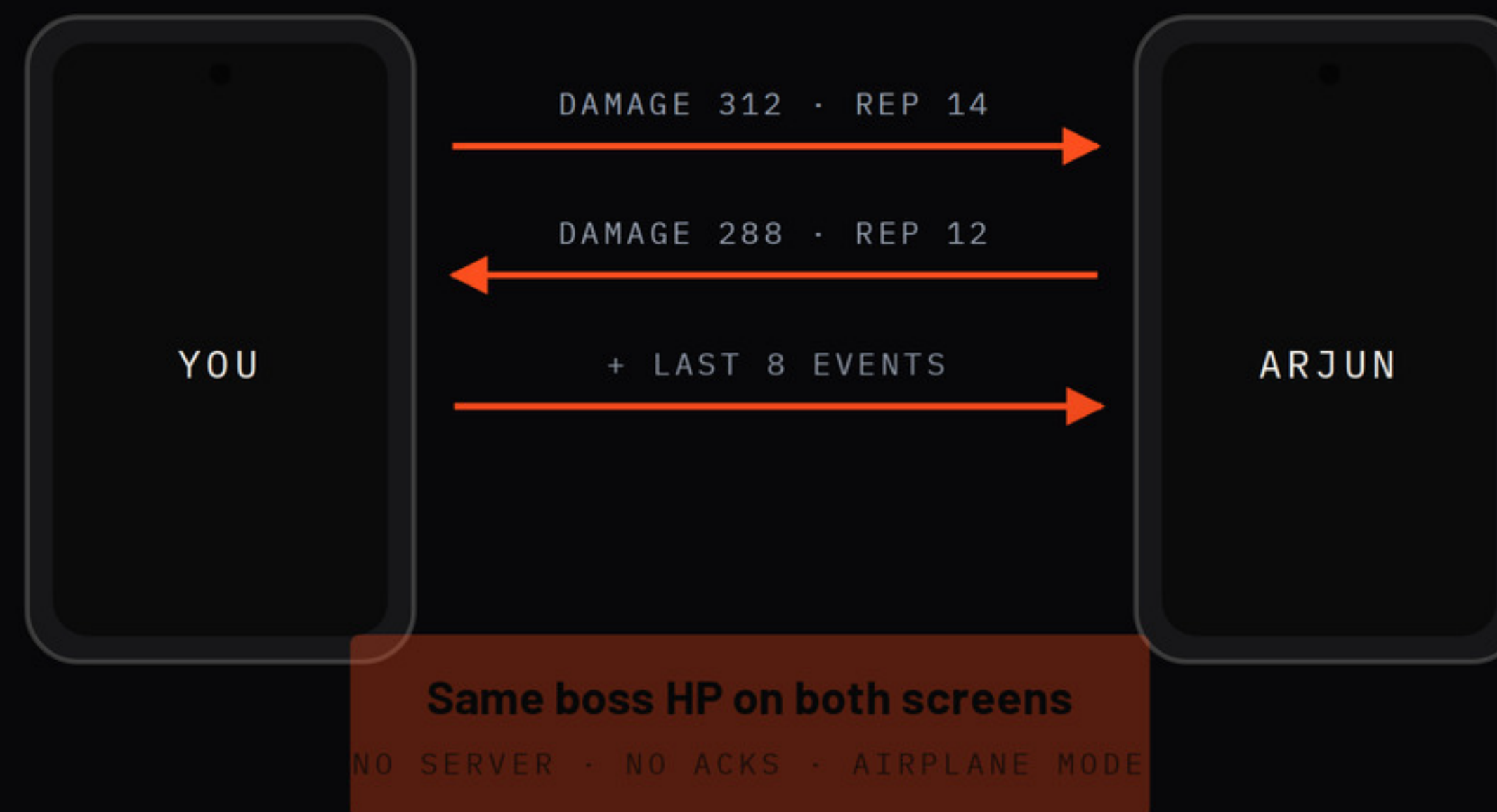
DUEL · MOST DAMAGE WINS

RAID · 3-8 CO-OP

LAST STANDING · FATIGUE KO

PASS THE PHONE · ONE DEVICE

EVENTS, NOT STATE



ONE ENGINE. FIFTEEN WAYS IN.

Every mode runs on the same camera, the same rep machine and the same fatigue estimator. Different rules over one honest stream of reps — which is why fifteen modes cost far less than fifteen games.

ALONE		HEAD TO HEAD		A ROOM	
Boss Fight	2-4 min	Duel	Most damage	Raid	3-8 co-op
Time Attack	60 s	Siege	Holds	Last Standing	Fatigue KO
Survival	Waves	Pursuit	Cardio	Pass the Phone	One device
Boss Rush	×3	Breaker	Jumps	Circuit	Four families
Tempo Trial	Eyes-free	Sigil	Yoga		
Ghost Race	Vs a file				

EVERY MODE NAMED HERE RESOLVES TO CODE IN CONFIG/COMBAT.JSON OR SRC/ROSTER.JS. A TEST FAILS THE BUILD IF ONE DOES NOT.

FOUR OF THESE FIVE NEVER INVOLVE A BOSS.

The camera that referees a fight is only a camera that can see a body. The same pipeline powers a set of health utilities whose price is verified movement — and none of them asks you to wear anything.

#	TOOL	WHAT IT DOES	REQUIRES
01	The alarm you cannot dismiss from bed	Set the alarm, then set its price: five push-ups, a thirty-second plank. It rings until the camera has counted them. An emergency dismiss is always there, held five seconds and logged.	Native app
02	A daily budget for the app you keep opening	Past the budget it stays shut until you pay in reps. Deletable in one tap on any day — a lock you cannot escape is a trap, not a feature.	Native app
03	Sixty seconds, every fifty minutes	A desk timer for the eight hours nobody trains in. The smallest honest unit of movement, verified rather than assumed.	Camera only
04	What your neck did all day	One frame every few minutes, scored for neck flexion and shoulder height, then discarded. A number at the end of the day; nothing stored, nothing diagnosed.	Camera only
05	Breathing you do not have to be trusted on	Six breaths a minute, confirmed from chest and shoulder motion, reporting the rate you actually held rather than the one you were asked for.	Camera only

THIS IS NOT A MOCKUP. IT ALREADY RUNS.

A working engine exists today, written by two people, with no runtime dependencies at all. The fatigue curve on slide 8 is not a drawing — it is that engine's output, and one command reproduces every number on it.

```
$ node test/run.js
160 passed, 0 failed

$ node test/replay.js traces/...
13 reps · fatigue 0.63 · GASSED · MERCY FIRED
```

Tests cover the rep machine, the form scorer, the fatigue estimator, the duel sync, the coach's validator and the landing page's own claims — the build fails if the page prints a number the config does not hold.

160

TESTS PASSING

31

MODULES

5 032

LINES OF ENGINE

0

RUNTIME DEPENDENCIES

51

EXERCISES IN CONFIG

15

MODES WIRED

Fifteen defects the suite caught before a human did — including a calibration step that took 27 seconds and ate six reps of a real set, a duel handshake that answered HELLO with HELLO until the stack overflowed, and a jump-height measurement that was structurally impossible because world landmarks are hip-centred.

— WHO THIS IS FOR

THREE PEOPLE, ONE ENGINE.

01 · THE WORKING PROFESSIONAL

Eight hours seated, forty minutes free

No commute to a gym, no equipment, no audience. A boss fight is two to four minutes and it fits between a meeting and dinner. The desk timer and the alarm reach the hours training never does.

OUR OWN BUCKET

02 · THE PERSON WHO TRAINS ALONE

Nobody has ever watched their form

Home training has no coach and no mirror worth trusting. Per-rep depth, range, tempo and alignment is feedback they have never had access to at any price they were willing to pay.

THE CORE LOOP

03 · THE PERSON IN REHAB

Two appointments, five months apart

A validated sit-to-stand and a symmetry index, measured at home between visits, produce the trend a twice-a-year assessment cannot. The clinic did not need a new product — it needed the one the game already built.

WHERE IT GOES NEXT

WHAT IS BUILT, AND WHAT IS HONESTLY NOT.

RUNNING TODAY

Rep detection across five movement families

Four-part form scoring with per-exercise weights

Fatigue estimator, four bands, five signal sets

Fatigue-adaptive boss with mercy resolution

Bilateral asymmetry with confidence grading

Two-phone duel over offline event sync

On-device coach with a hallucination validator

30-second sit-to-stand assessment

NEEDS THE NATIVE APP

Wake-up alarm – a browser cannot fire one

App-unlock tax – needs usage-access permission

Background posture sampling

Run tracker with an on-device route

FURTHER OUT

Group raid rooms beyond two devices

Vision-grounded critique on the worst rep

Physio-prescribed protocols by share code

The full clinic battery – five validated tests

WE SEPARATE THESE DELIBERATELY. A TEAM THAT BLURS SHIPPED AND PLANNED GETS CAUGHT IN THE DEMO; A TEAM THAT SEPARATES THEM CLEANLY IS TELLING YOU WHAT ITS JUDGEMENT IS LIKE.

— THE TEAM

DA GOATS. TWO WORKING ENGINEERS.

Omkar Kadam

ANDROID & PERCEPTION

Owens the camera path, the pose pipeline and the detector families — everything between a frame arriving and a rep being counted.

Ujjwal Pardeshi

GAME SYSTEMS & ON-DEVICE INFERENCE

Owens the combat model, the multiplayer sync and the on-device language model — everything between a rep being counted and it meaning something.

WORKING PROFESSIONALS

TEAM OF TWO

PUNE CITY BATTLE

BUILDING THIS SINCE BEFORE THE HACKATHON

— CLASHFIT · TEAM DA GOATS

PROP YOUR PHONE. FIGHT IT OUT.

Real, camera-verified effort in. Boss damage out. Every rep graded on depth, range, tempo and alignment — and none of it ever leaves your phone.

`clash-fit.vercel.app`